Course: Cryptography and Network Security
Code: CSN15403
Branch: B.Tech -5th Semester

Lecture – 8 : Data Encryption Standard (DES)

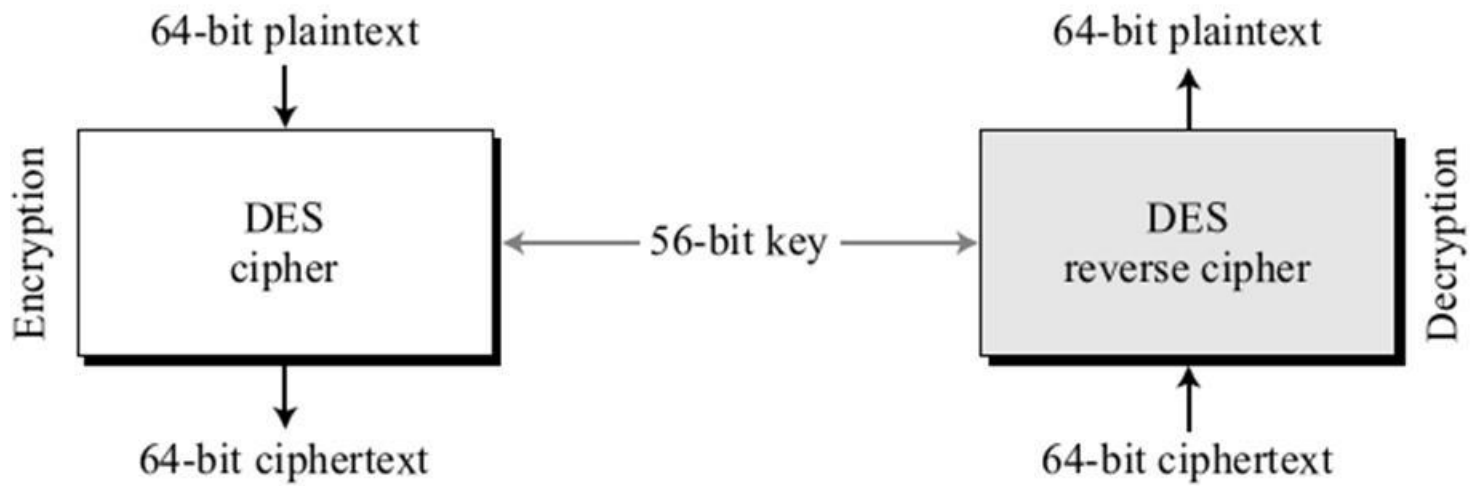
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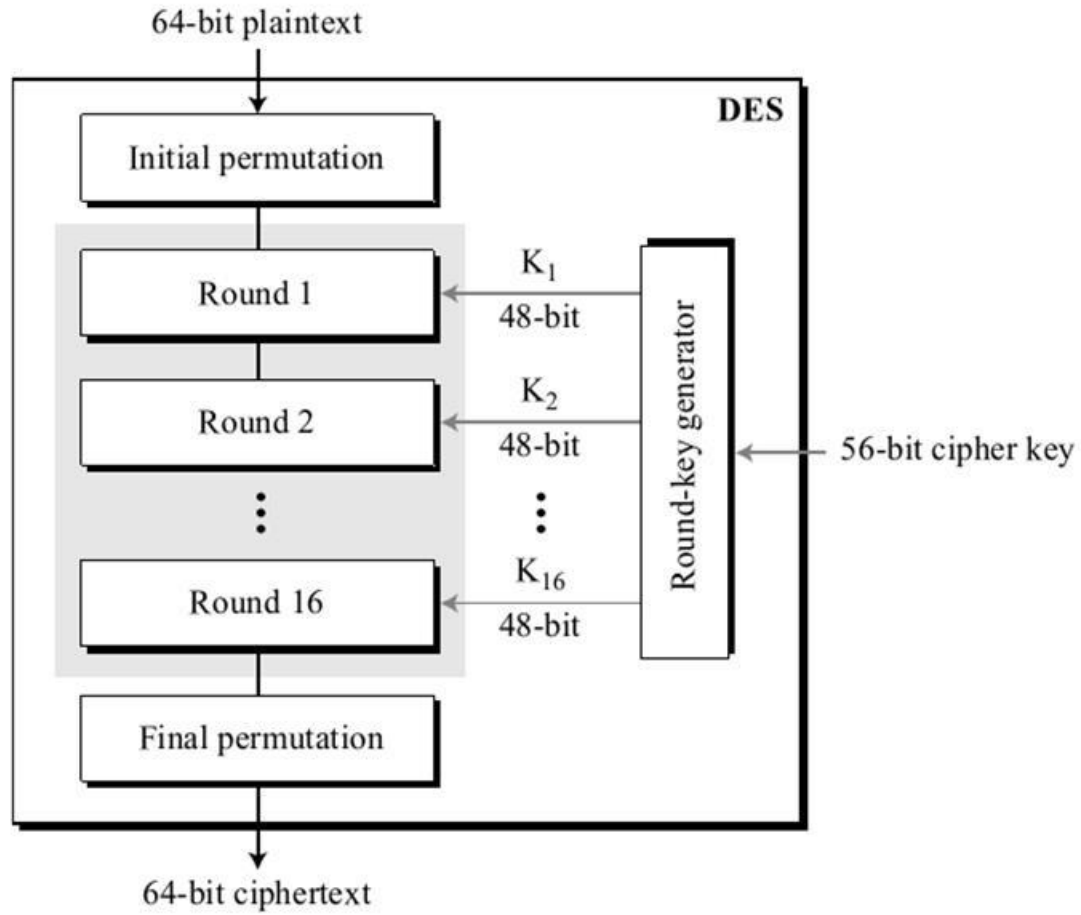
INTRODUCTION

- The **Data Encryption Standard (DES)** is a symmetric-key block cipher published by the **National Institute of Standards and Technology (NIST).**
- DES is a block cipher

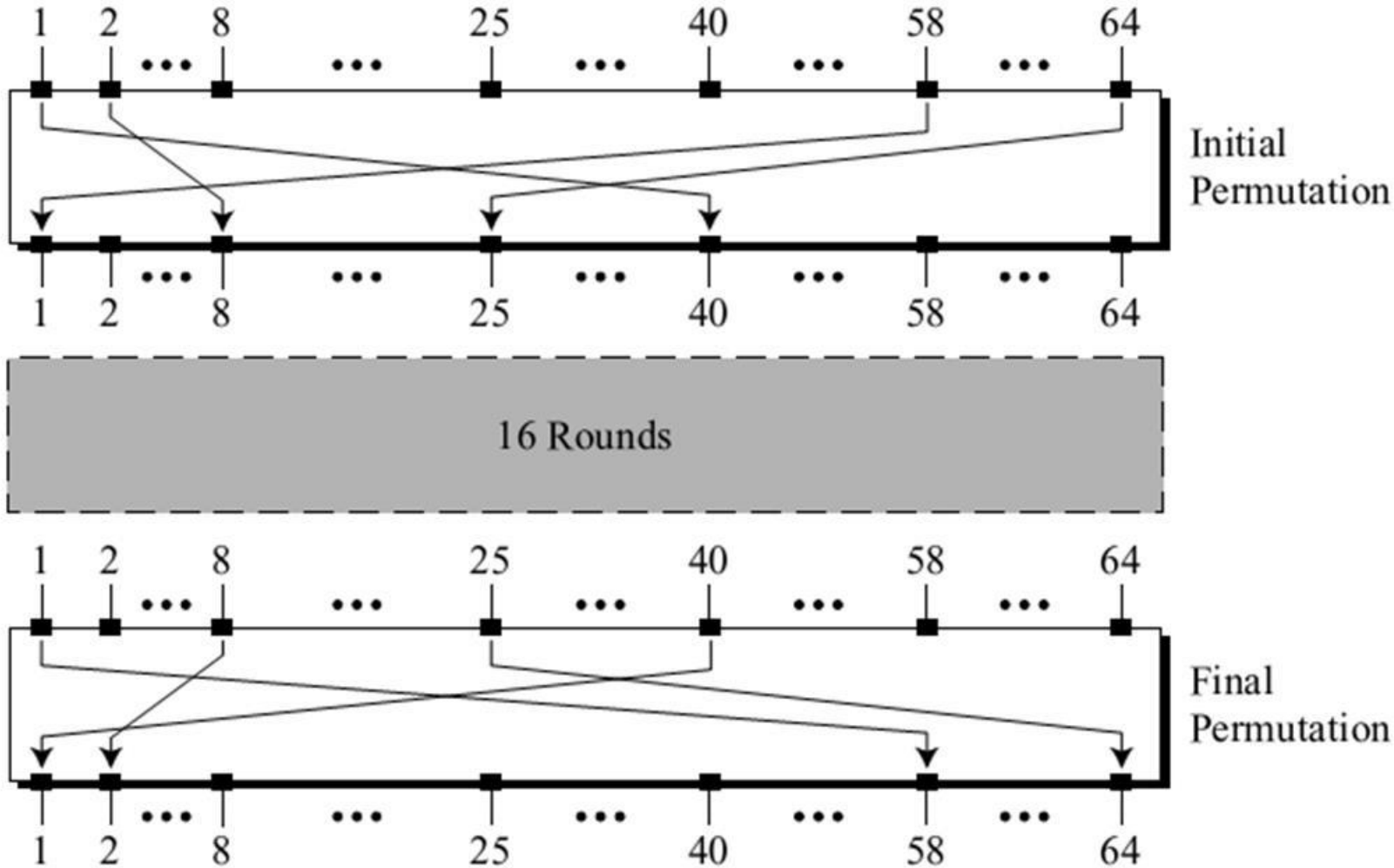


DES STRUCTURE

- The encryption process is made of two permutations (P-boxes), which we call initial and final permutations, and sixteen Feistel rounds.
- Each round uses a different 48-bit round key generated from the cipher key according to a predefined algorithm
- The initial and final permutations are straight P-boxes that are inverses of each other.
- They have no cryptography significance in DES.



Initial and Final Permutations

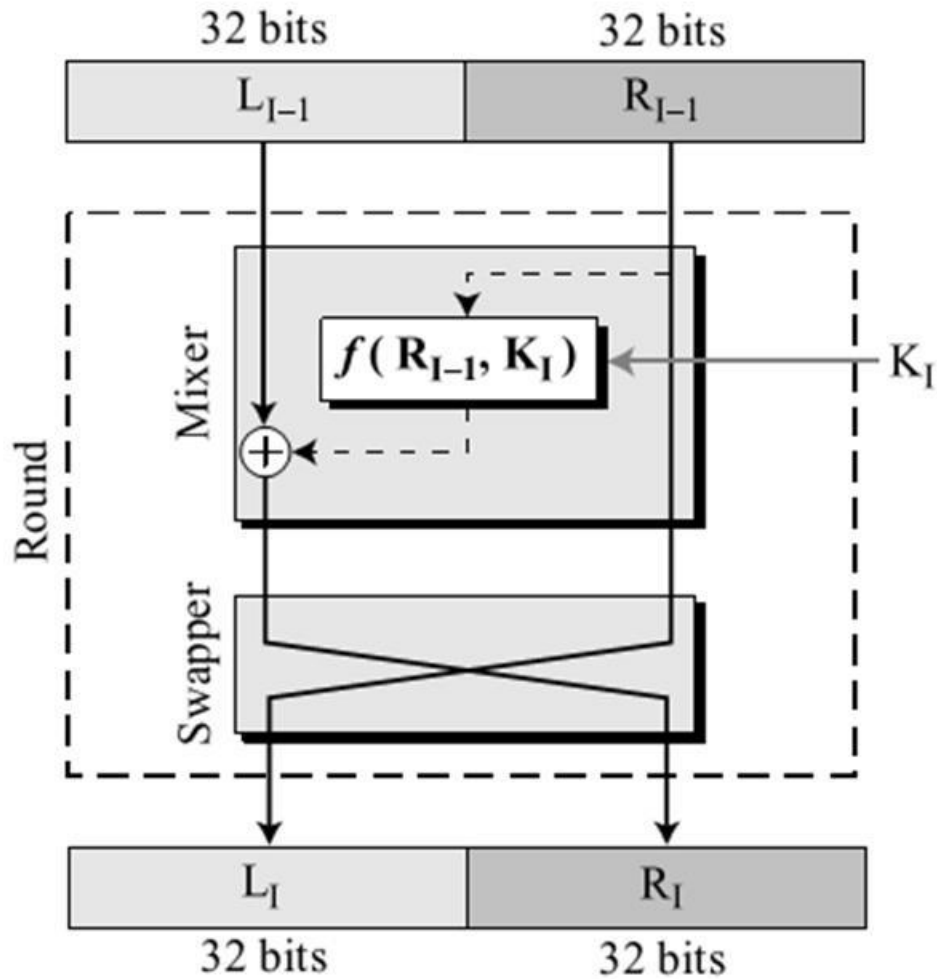


Initial and final permutation tables

Initial Permutation								*Final Permutation*							
58	50	42	34	26	18	10	02	40	08	48	16	56	24	64	32
60	52	44	36	28	20	12	04	39	07	47	15	55	23	63	31
62	54	46	38	30	22	14	06	38	06	46	14	54	22	62	30
64	56	48	40	32	24	16	08	37	05	45	13	53	21	61	29
57	49	41	33	25	17	09	01	36	04	44	12	52	20	60	28
59	51	43	35	27	19	11	03	35	03	43	11	51	19	59	27
61	53	45	37	29	21	13	05	34	02	42	10	50	18	58	26
63	55	47	39	31	23	15	07	33	01	41	09	49	17	57	25

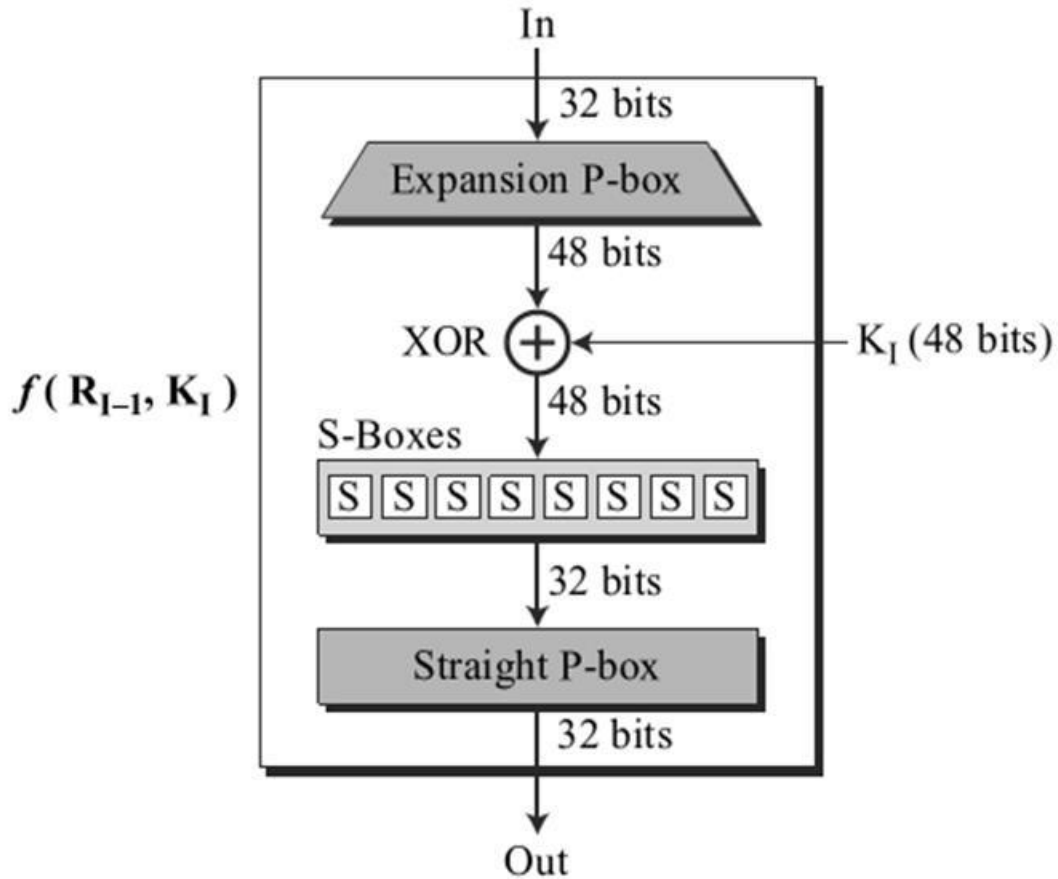
Rounds

- DES uses 16 rounds.
- Each round of DES is a Feistel cipher
- The round takes L_{I-1} and R_{I-1} from previous round (or the initial permutation box) and creates L_I and R_I, which go to the next round (or final permutation box).
- The mixer is invertible because of the XOR operation.

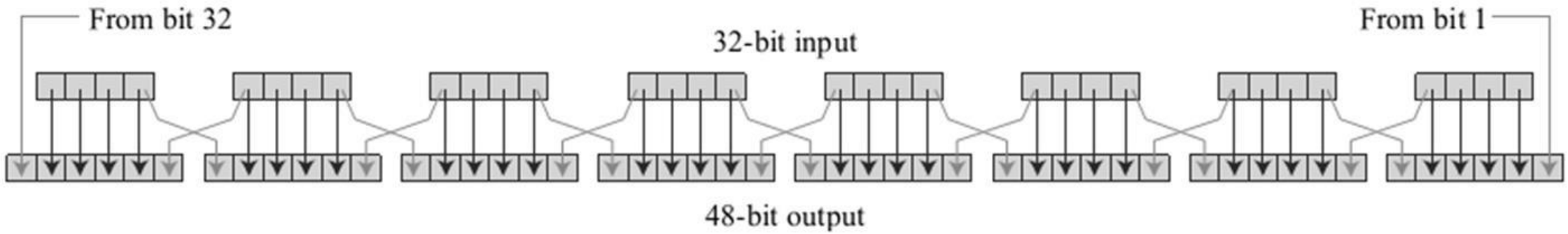


DES Function

- The heart of DES is the DES function.
- The DES function applies a 48-bit key to the rightmost 32 bits (R_{I-1}) to produce a 32-bit output.
- This function is made up of four sections:
  - an expansion P-box,
  - a whitener (that adds key),
  - a group of S-boxes, and
  - a straight P-box



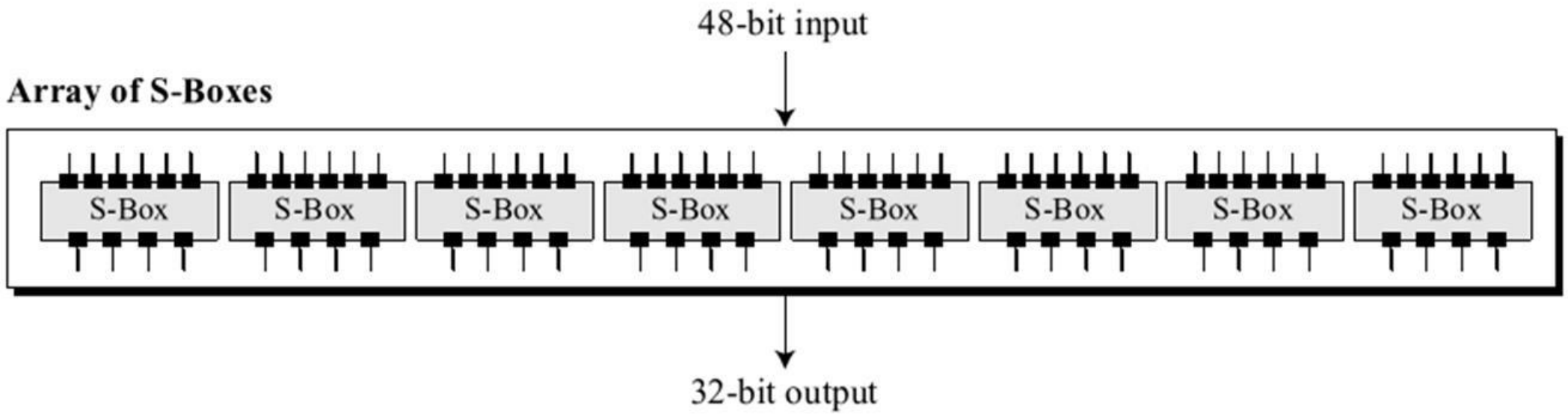
Expansion permutation and Expansion P-box table



32	01	02	03	04	05
04	05	06	07	08	09
08	09	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	01

S-Boxes

The S-boxes do the real mixing (confusion). DES uses 8 S-boxes, each with a 6-bit input and a 4-bit output.



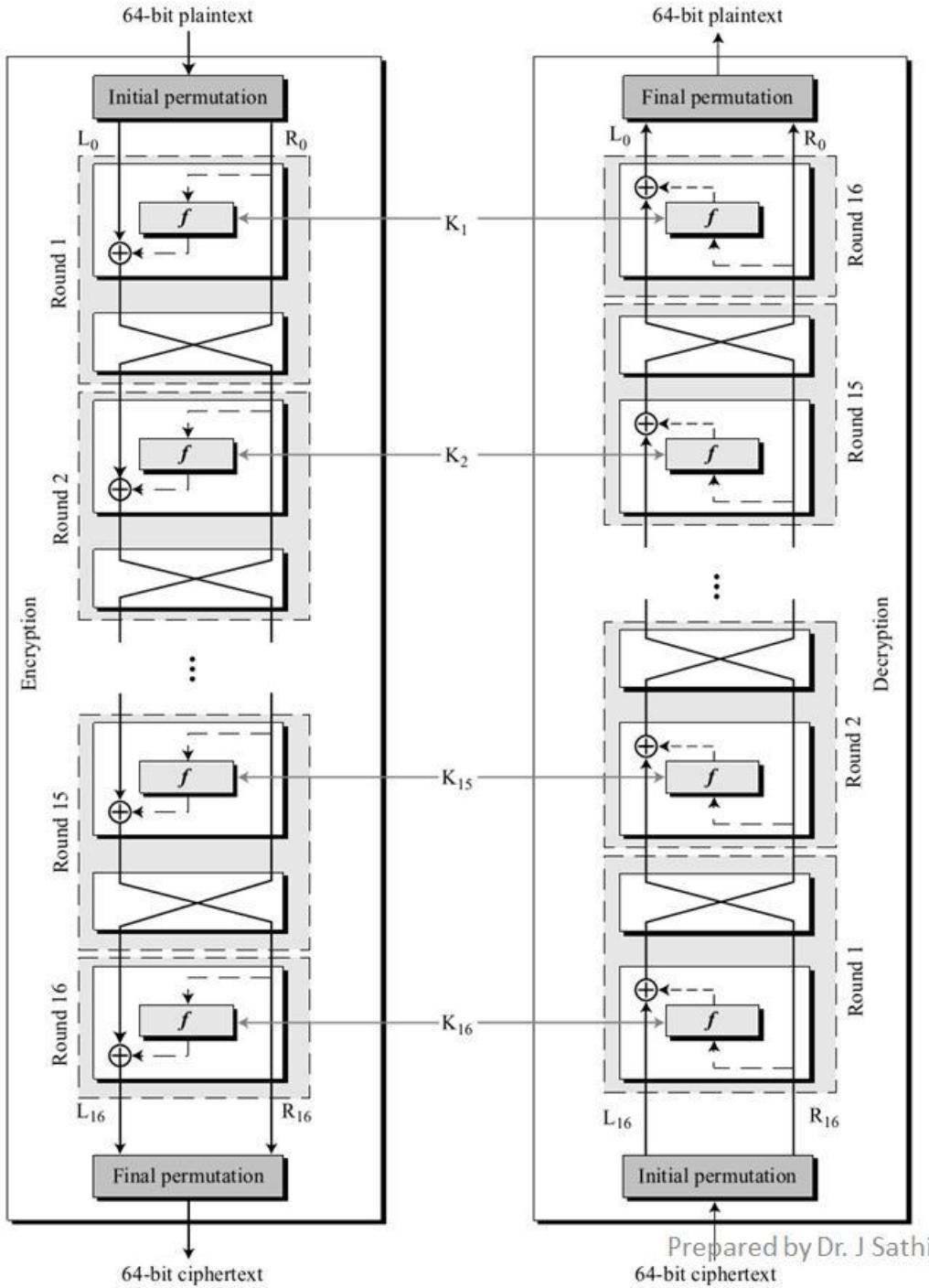
Straight Permutation

- The last operation in the DES function is a straight permutation with a 32-bit input and a 32-bit output
- The seventh bit of the input becomes the second bit of the output

16	07	20	21	29	12	28	17
01	15	23	26	05	18	31	10
02	08	24	14	32	27	03	09
19	13	30	06	22	11	04	25

Cipher and Reverse Cipher

- Using mixers and swappers, we can create the cipher and reverse cipher, each having 16 rounds.
- The cipher is used at the encryption site; the reverse cipher is used at the decryption site.
- The whole idea is to make the cipher and the reverse cipher algorithms similar.
- Many Approaches to Implement DES



DES cipher and reverse cipher for the first approach

Pseudocode for DES cipher

```
Cipher (plainBlock[64], RoundKeys[16, 48], cipherBlock[64])
{
    permute (64, 64, plainBlock, inBlock, InitialPermutationTable)
    split (64, 32, inBlock, leftBlock, rightBlock)
     for (round = 1 to 16)
     {
          mixer (leftBlock, rightBlock, RoundKeys[round])
           if (round!=16) swapper (leftBlock, rightBlock)
    }
     combine (32, 64, leftBlock, rightBlock, outBlock)
     permute (64, 64, outBlock, cipherBlock, FinalPermutationTable)
}
```

Pseudocode for DES cipher

```
mixer (leftBlock[32], rightBlock[32], RoundKey[48])
{
    copy (32, rightBlock, T1)
    function (T1, RoundKey, T2)
    exclusiveOr (32, leftBlock, T2, T3)
    copy (32, T3, rightBlock)
}
```

```
swapper (leftBlock[32], rigthBlock[32])
{
    copy (32, leftBlock, T)
    copy (32, rightBlock, leftBlock)
    copy (32, T, rightBlock)
}
```

Pseudocode for DES cipher

```
function (inBlock[32], RoundKey[48], outBlock[32])
{
    permute (32, 48, inBlock, T1, ExpansionPermutationTable)
    exclusiveOr (48, T1, RoundKey, T2)
    substitute (T2, T3, SubstituteTables)
    permute (32, 32, T3, outBlock, StraightPermutationTable)
}
```

Pseudocode for DES cipher

```
substitute (inBlock[32], outBlock[48], SubstitutionTables[8, 4, 16])
{
    for (i = 1 to 8)
    {
        row ← 2 × inBlock[i × 6 + 1] + inBlock [i × 6 + 6]
        col ← 8 × inBlock[i × 6 + 2] + 4 × inBlock[i × 6 + 3] +
              2 × inBlock[i × 6 + 4] + inBlock[i × 6 + 5]

        value = SubstitutionTables [i][row][col]

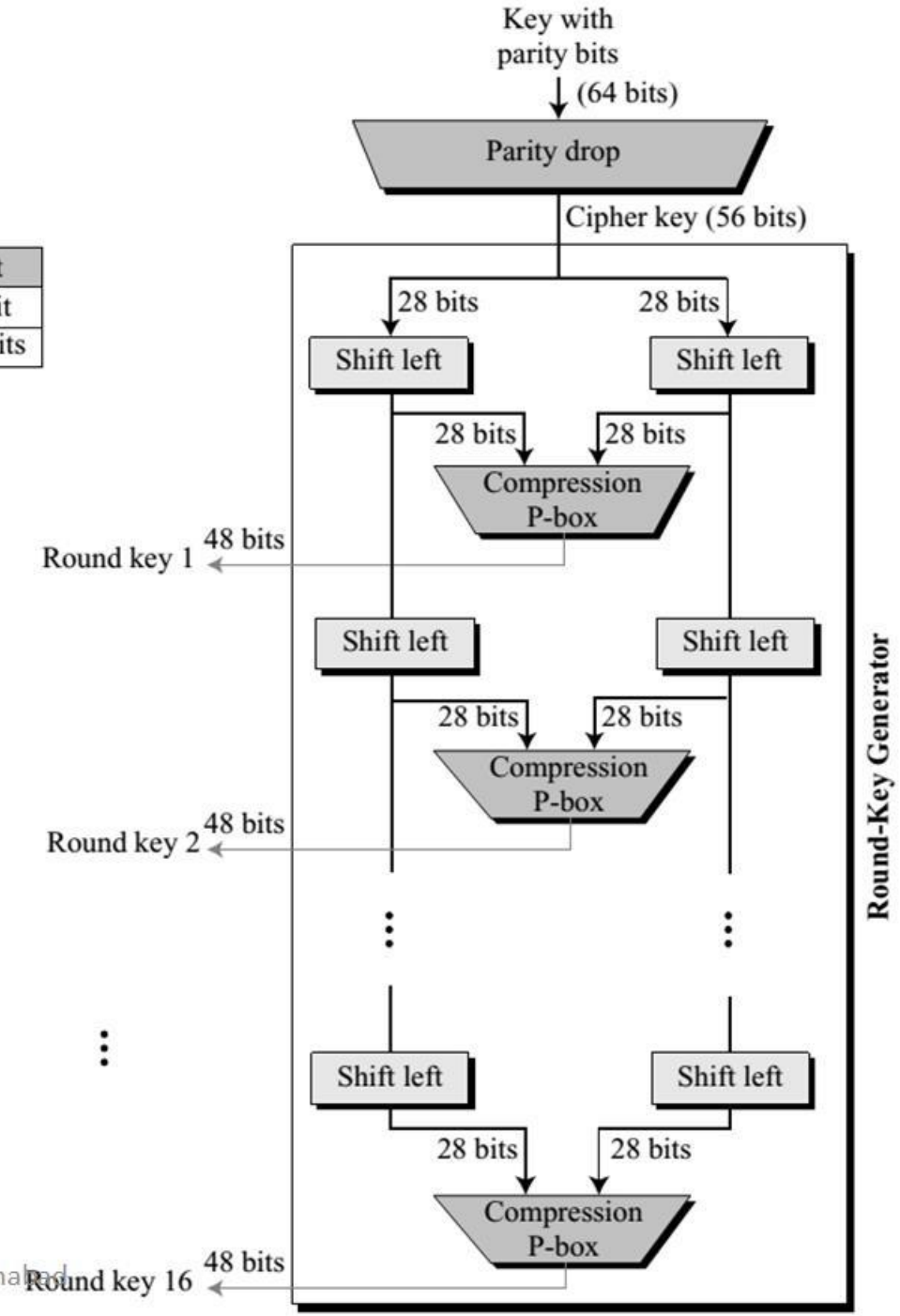
        outBlock[[i × 4 + 1] ← value / 8;      value ← value mod 8
        outBlock[[i × 4 + 2] ← value / 4;      value ← value mod 4
        outBlock[[i × 4 + 3] ← value / 2;      value ← value mod 2
        outBlock[[i × 4 + 4] ← value
    }
}
```

Key Generation

- The round-key generator creates sixteen 48-bit keys out of a 56-bit cipher key.
- However, the cipher key is normally given as a 64-bit key in which 8 extra bits are the parity bits, which are dropped before the actual key-generation process.

Shifting

Rounds	Shift
1, 2, 9, 16	one bit
Others	two bits



Algorithm for round-keys generation

```
Key_Generator (keyWithParities[64], RoundKeys[16, 48], ShiftTable[16])
{
    permute (64, 56, keyWithParities, cipherKey, ParityDropTable)
    split (56, 28, cipherKey, leftKey, rightKey)
     for (round = 1 to 16)
     {
          shiftLeft (leftKey, ShiftTable[round])
          shiftLeft (rightKey, ShiftTable[round])
          combine (28, 56, leftKey, rightKey, preRoundKey)
          permute (56, 48, preRoundKey, RoundKeys[round], KeyCompressionTable)
     }
}
```

DES ANALYSIS

- Critics have used a strong magnifier to analyze DES.
- Tests have been done to measure the strength of some desired properties in a block cipher.
- Properties
  - Two desired properties of a block cipher are the avalanche effect and the completeness.
- Avalanche Effect
  - Avalanche effect means a small change in the plaintext (or key) should create a significant change in the ciphertext.
  - DES has been proved to be strong with regard to this property.
- Completeness effect
  - Completeness effect means that each bit of the ciphertext needs to depend on many bits on the plaintext.
  - The diffusion and confusion produced by P-boxes and S-boxes in DES, show a very strong completeness effect.

DES Weaknesses

- Weaknesses in Cipher Design
  - S-boxes
    - In S-box 4, the last three output bits can be derived in the same way as the first output bit by complementing some of the input bits.
    - Two specifically chosen inputs to an S-box array can create the same output.
    - It is possible to obtain the same output in a single round by changing bits in only three neighboring S-boxes.
  - P-boxes
    - It is not clear why the designers of DES used the initial and final permutations; these have no security benefits.
    - In the expansion permutation (inside the function), the first and fourth bits of every 4-bit series are repeated.

DES Weaknesses

- Weakness in the Cipher Key
  - DES is in its key size (56 bits). To do a brute-force attack on a given ciphertext block, the adversary needs to check 2^{56} keys.
  - With available technology, it is possible to check one million keys per second.
  - If we can make a computer with one million chips (parallel processing), then we can test the whole key domain in approximately 20 hours.
  - Computer networks can simulate parallel processing. In 1977 a team of researchers used 3500 computers attached to the Internet to find a key challenged by RSA Laboratories in 120 days. The key domain was divided among all of these computers, and each computer was responsible to check the part of the domain.
  - If 3500 networked computers can find the key in 120 days, a secret society with 42,000 members can find the key in 10 days

DES Weaknesses

- Weak Keys
  - Four out of 2^{56} possible keys are called weak keys
  - The round keys created from any of these weak keys are the same and have the same pattern as the cipher key.

Keys before parities drop (64 bits)	*Actual key (56 bits)*
0101 0101 0101 0101	0000000 0000000
1F1F 1F1F 0E0E 0E0E	0000000 FFFFFFF
E0E0 E0E0 F1F1 F1F1	FFFFFFF 0000000
FEFE FEFE FEFE FEFE	FFFFFFF FFFFFFF

DES Weaknesses

- Semi-weak Keys
  - There are six key pairs that are called semi-weak keys.
  - A semi-weak key creates only two different round keys and each of them is repeated eight times.

First key in the pair	*Second key in the pair*
01FE 01FE 01FE 01FE	FE01 FE01 FE01 FE01
1FE0 1FE0 0EF1 0EF1	E01F E01F F10E F10E
01E0 01E1 01F1 01F1	E001 E001 F101 F101
1FFE 1FFE 0EFE 0EFE	FE1F FE1F FE0E FE0E
011F 011F 010E 010E	1F01 1F01 0E01 0E01
E0FE E0FE F1FE F1FE	FEE0 FEE0 FEF1 FEF1

DES Weaknesses

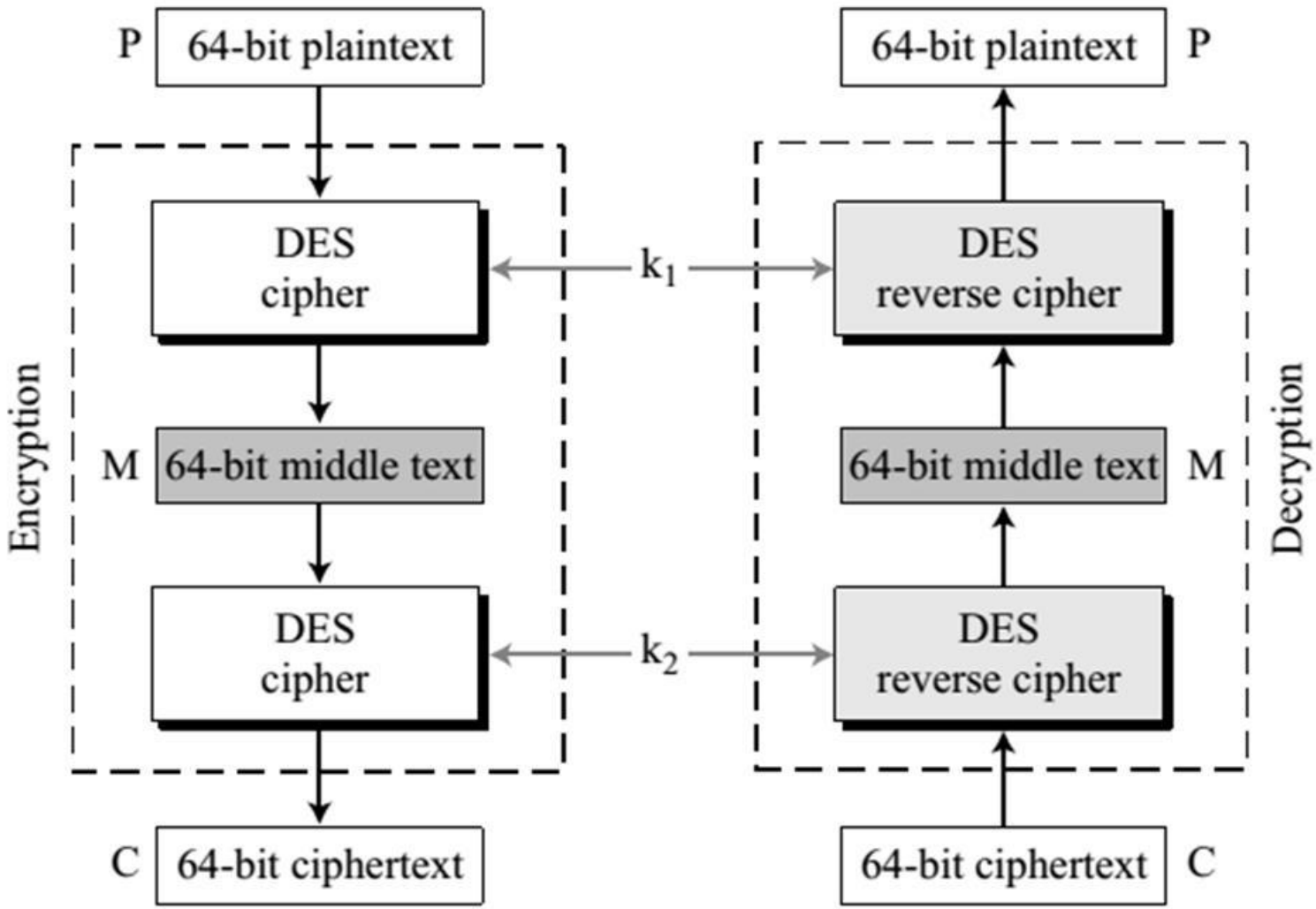
- Semi-weak Keys
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First key in the pair	*Second key in the pair*
01FE 01FE 01FE 01FE	FE01 FE01 FE01 FE01
1FE0 1FE0 0EF1 0EF1	E01F E01F F10E F10E
01E0 01E1 01F1 01F1	E001 E001 F101 F101
1FFE 1FFE 0EFE 0EFE	FE1F FE1F FE0E FE0E
011F 011F 010E 010E	1F01 1F01 0E01 0E01
E0FE E0FE F1FE F1FE	FEE0 FEE0 FEF1 FEF1

MULTIPLE DES

- The major criticism of DES regards its key length.
- With available technology and the possibility of parallel processing, a brute-force attack on DES is feasible.
- One solution to improve the security of DES is to abandon DES and design a new cipher, with the advent of AES.
- The second solution is to use multiple (cascaded) instances of DES with multiple keys
  - Double DES
  - Triple DES

Double DES


P
64-bit plaintext
DES cipher
k_1
M
64-bit middle text
DES cipher
k_2
C
64-bit ciphertext
Encryption
64-bit plaintext
P
DES reverse cipher
64-bit middle text
M
DES reverse cipher
64-bit ciphertext
C
Decryption

Meet-in-the-Middle Attack

- Using a known-plaintext attack called meet-in-the-middle attack proves that double DES improves this vulnerability slightly (to 2^{57} tests), but not tremendously (to 2^{112}).

$$M = E_{k_1}(P) \qquad \text{and} \qquad M = D_{k_2}(C)$$

- Eve encrypts P using all possible values (2^{56}) of k1 and records all values obtained for M.
- Eve decrypts C using all possible values (2^{56}) of k2 and records all values obtained for M.
- This means that instead of using 2^{112} key-search tests, Eve uses 2^{56} key-search tests two times

Triple DES

- To improve the security of DES, triple DES (3DES) was proposed.
- This uses three stages of DES for encryption and decryption.
- Two versions of triple DES are in use today:
  - triple DES with two keys and
  - triple DES with three keys.

Triple DES with two keys

